Empirical Multiplier Summary

SubXPAT experiment snapshot

March 31, 2026

Scope

This summary covers the two confirmed multiplier sweeps that were rebuilt after cleanup:

- 8-bit consistency rerun
- 10-bit rerun

The 12-bit sweep was still running when this document was generated, so it is not included here. Runtime values are wall-clock times from the batch launcher on a shared server, so broad gaps are meaningful while 1–5 second differences should be treated as noise.

Best Settings So Far

Width	Best tested setting	Final area	Runtime (s)	Why it stands out
8-bit	`none NINE e60 a2 b30`	2.8158	42	Ties the minimum area and is the fastest among the minimum-area 8-bit settings.
10-bit	`none NINE e60 a4 b10`	5.6316	118	Best 10-bit result in both area and runtime within the tested set.

Termination Mode Takeaways

Width	Case	Δ area (trivial - none)	Δ time (s)	Interpretation
8-bit	`NINE e15 a1 b80`	+2.3465	-252	Trivial is much faster, but it gives up some area quality.
8-bit	`NINE_PRIME e15 a2 b30 c5`	0.0000	-90	Trivial is clearly better here because it saves time with no area loss.
10-bit	`NINE e60 a1 b80`	-0.4690	-444	Trivial wins on both runtime and area in this case.
10-bit	`NINE_PRIME e60 a2 b30 c5`	+2.3470	-656	Trivial is dramatically faster, but none keeps the better area.

Main Conclusions

- The strongest default direction so far is the `NINE` family, not `NINE_PRIME`.

- The `a4 b10` setting is the standout 10-bit choice. On 8-bit, the best area can be reached by multiple settings, but `e60 a2 b30` is the fastest of those ties.
- `trivial` termination is useful when the goal is faster sweeps, but it is not uniformly safe: it helps a lot on some cases and noticeably hurts area on others.
- The `a1 b80` high-beta setting is not a good general default. It remains weak on 8-bit and only becomes attractive on 10-bit because trivial termination collapses the runtime.

Most Useful Visuals

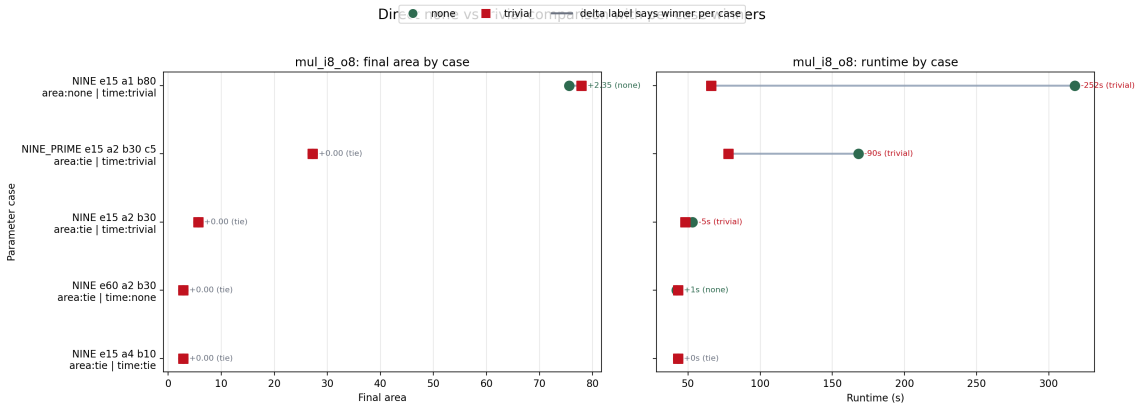

Figure 1: 8-bit direct comparison of `none` vs. `trivial` termination for each tested case.

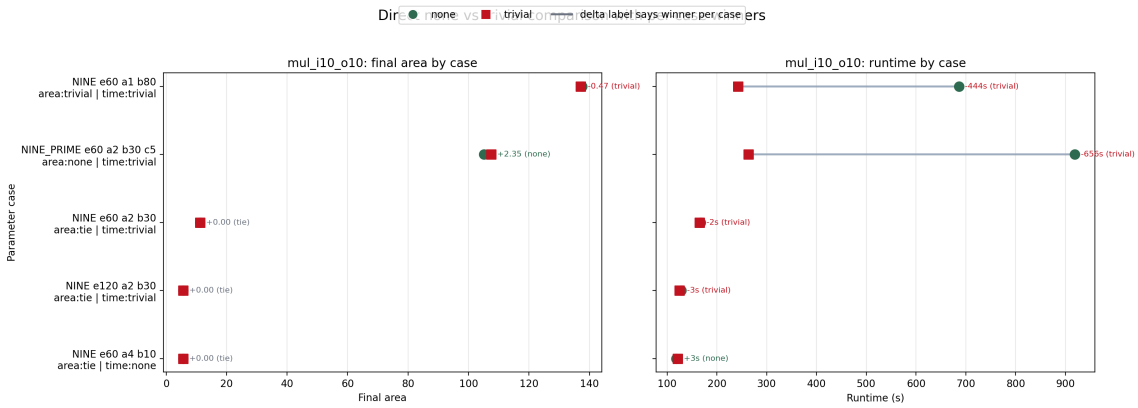

Figure 2: 10-bit direct comparison of `none` vs. `trivial` termination for each tested case.

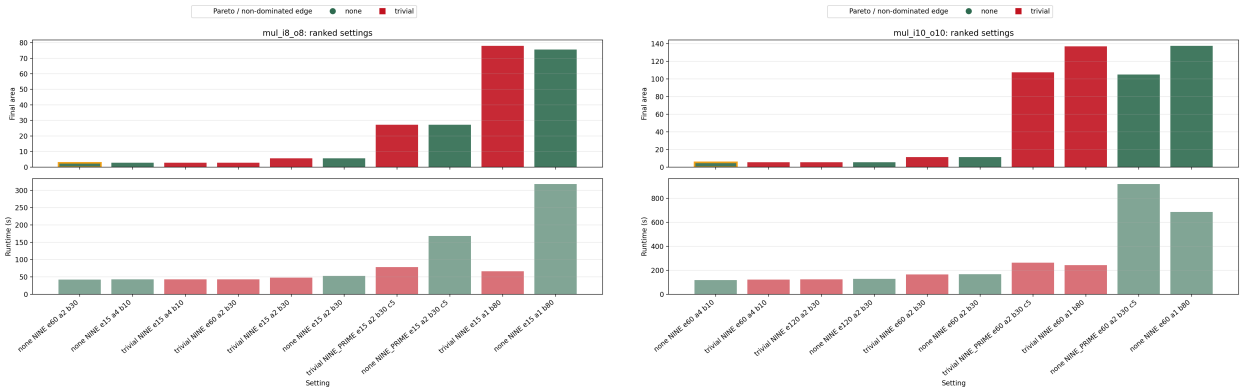

Figure 3: Ranked parameter scorecards for the confirmed 8-bit and 10-bit sweeps.

Source Folders

- `benchmarking/generated/multiplier_empirical_8bit_consistency_20260331`
- `benchmarking/generated/multiplier_empirical_10bit_rerun_20260331`
- `output/figure/multiplier_empirical_8bit_consistency_20260331`
- `output/figure/multiplier_empirical_10bit_rerun_20260331`